

Memorial University of Newfoundland
CMSC 6950: Computer Based Research Tools and Applications
Computer Tools – Fall 2025

In this class, we will use

- the Unix shell, Bash,
- Python,
- Jupyter notebooks,
- Git,
- and various packages/extensions to the above.

You have three options for how you use these tools:

1. on your own computer (that you can bring to class),
2. on the LabNet computers in the classroom,
3. or on another machine that you can access remotely from the classroom.

If you are using your own computer, it is your responsibility to ensure that you have functioning tools. You may wish to consult instructions for setting things up, such as

- <https://object-oriented-python.github.io/installation.html>
- https://carpentries.github.io/workshop-template/install_instructions

You are welcome to use Anaconda in this course to provide Python tools, but be aware that Anaconda manages Python dependencies in a way that is difficult to use when you must interact with several packages at once.

On LabNet machines, you should have everything you need once you boot into Linux mode. If you do not have a LabNet account yet, see the information at <https://www.labnet.mun.ca>.

If you wish to connect remotely to another computer, please be sure to have appropriate connection tools on your device or know how to use those from the LabNet machines in the classroom. In this class, we'll use ssh. If you use Windows, consider installing Putty or MobaXterm on your machine to enable remote connections.